Kaustubh Agarwal

ML models, evaluation, external validation, the close

Run order: slides 6–8, 11–12, 16–17 + demo

NUMBERS COLD

- Behaviour **91.6 %** — *say "on synthetic labels" in the same breath*
- Chat **PR-AUC 0.825**; at the 0.95 alert threshold **P 0.956** / **R 0.428**
- Voice **0.574** speaker-independent (random splits said 0.657 — an 8.3-point leakage we found and corrected)
- ρ **= 0.317** [0.137, 0.478], n = 103 — the pre-designated primary endpoint
- Screen-time baseline **0.147**; model ahead in **97.4 %** of resamples ($\Delta\rho$ = +0.167, CI grazes zero — *say "suggestive, not established"*)
- **Partial** ρ **= 0.303** [0.110, 0.476] — hours removed, still excludes zero
- **Pattern** $-$ **volume = +0.199 [+0.019, +0.380]** — formally tested, excludes zero. *This is your strongest number. Reach for it whenever the $\Delta\rho$ is attacked.*
- Genre p = 0.598 (32 % power) · 1 disordered respondent → ~157 usable needed for caseness
- Svarah fairness: **0 false alerts** / **9.6 h**; WER 35 % → 61 % across L1 families

HARDEST QUESTIONS

"0.32 is a weak correlation."
~10 % shared variance, and 0.3–0.5 is the normal band for a brief screener against a full instrument (PHQ-2 vs PHQ-9). We claim construct carriage, not clinical accuracy. Then pivot to the baseline: the thing it beats is the screen-time counter every shipping tool uses.

"Why no sensitivity / specificity?"
One respondent scored in the disordered range. The guard that refuses those metrics below a minimum positive count was written *before* the data arrived. Reporting them off n=1 would be the dishonest choice.

"Your training labels are synthetic."
Correct, and stated on every slide it matters. That is exactly why the survey exists — it is where the served score meets a real clinical instrument.

"Your headline dropped from 0.352 to 0.317."
Say it before they find it. A late batch of 22 arrived after the main wave; the rule for including late responses was fixed before any of it existed, so we folded it in and reported the attenuated number.

"Why is the genre multiplier still there?"
An underpowered null is not evidence of no effect (32 % power). We flag it, we publish the failed test, and we report that removing it flips 34 % of bands — which is precisely why it needs a powered study, not a silent deletion.

Hand-offs: device/battery numbers → Khushee · thresholds & fusion → Kanak · ethics framing → Vidisha. **Protect slide 12's time.** Never say "addicted" or "diagnosis".

Khushee P Kiran

Android apps, passive capture, anti-tamper, device cost **Run order: slides 4, 10 (capture half), 14 — and you drive the demo**

NUMBERS COLD

- Default path (English STT): **14 % of one core, 288 MB mean / 301 MB peak** — passes every gate
- Dual Hindi+English STT: **51–72 % CPU, 399/419 MB — fails 2 of our 3 gates, which is why it ships default OFF**
- Measured 2026-08-18, Galaxy M52 5G, **two back-to-back 15-min Roblox sessions**, per-minute sampling
- Data: **7.16 MB up / 0.19 MB down** per 15-min session · temp drift 37.5 → 39.2 °C · **no thermal throttling**
- Cold start **569 ms** true-cold, ∼230 ms warm · session auto-start ∼10 s, auto-end ∼25 s, nudge ∼12 s
- For scale: *Roblox itself* runs at 215 % CPU — our monitor is a rounding error next to the game
- v2.4.0 signed release; signing cert unchanged since day one

HARDEST QUESTIONS

"Did you actually measure battery, or are you guessing?"
Measured, and the honest footnote comes first: `batterystats` reports zero power while the phone is on USB for adb, so we report the charging-safe proxy Android's own model uses — averaged per-app CPU time — plus RAM, thermal and data. Full table in `TESTING.md`.

"Can a child just evade it?"
Yes, in one disclosed way: browser and webview games. Our boundary is *installed apps, not games*. Closing it needs screen-capture or URL surveillance, which we refuse — a disclosed boundary beats a privacy regression.

"What if they switch keyboards?"
Detected. The heartbeat carries capture-capability bits; the parent gets an alert and the child gets a self-heal prompt.

"Clear app data and the observation counter resets?"
No — the counter is server-side. And the heartbeat dying is itself the alert.

"Can adb uninstall it?"
We tried. `DELETE_FAILED_DEVICE_POLICY_MANAGER` — Device Admin blocks it, and disabling admin fires an instant parent alert.

Before the room: both phones charged, on venue Wi-Fi, backend pre-warmed (`/api/health` → `models_loaded: true`), video fallback on the laptop desktop. **Hand-offs:** model numbers → Kaustubh · server/CI → Kanak.

Kanak Goyal

Backend, deployment, fusion & alerting, dataset provenance **Run order: slides 5, 9, 13, 14 (backend half)**

NUMBERS COLD

- Fusion **40/30/30**, renormalised over whichever channels are actually present
- Chat aggregation is **max, not mean** — one severe message must not be averaged away
- Operating points: 0.85 → P 0.800/R 0.703 · 0.90 → 0.888/0.623 · **0.95** → **0.956/0.428 (deployed)**
- Feedback tuner: capped at ±**0.05**, floored, and **human-applied** — there is no automatic loop
- **180 backend tests** on SQLite *and* Postgres 16, + 110 JVM + 7 research-integrity guards
- Load: **288 concurrent requests, 0 errors, p50 66 ms** · 512 MB serving budget
- Datasets: **11 adopted, 2 rejected with written evidence**, + StudentLife as a reality check

HARDEST QUESTIONS

"Why threshold 0.95 instead of best-F1?"
Base-rate arithmetic. A false alarm costs parental trust, and trust is the product. At best-F1 roughly 1 in 5 alerts is wrong; at 0.95 it is about 1 in 23. We chose precision and we say what it costs us — recall drops to 0.428.

"Can an anxious parent drag the thresholds down until everything alerts?"
No. The tuner is capped at ±0.05, floored, and never applies itself — a human does. There is no closed feedback loop for exactly this reason.

"Why RF and logistic regression instead of BERT?"
Measured, not assumed: off-the-shelf toxic-BERT scores 0.709 on our split against our 0.825, inside a 512 MB serving budget. We report the comparison rather than claiming transformers were infeasible.

"Why did you reject those Kaggle datasets?"
We downloaded and inspected them. Synthetic provenance in one; the other measures engagement, which is not addiction. Both rejections are written up with the evidence — §7 of the notes.

"Where are the fusion weights validated?"
Straight answer: they are clinically-motivated priors, and the sensitivity analysis is in the paper — we show how much the served bands move as they change. A validated weighting needs the telemetry-linked cohort, which is stage two.

Hand-offs: validation statistics → Kaustubh · on-device cost → Khushee · consent/privacy → Vidisha.

Vidisha Murali

Ethics, privacy, scope, the India story — and you open

Run order: slides 2–3, 10 (chat/fairness half), 15

NUMBERS COLD

- Export scope **=** deletion scope, 15 tables, credentials excluded
- Consent is **versioned and fails closed** — 403 until re-consent
- **Raw audio is never retained**: STT on-device, features extracted, audio deleted
- Hindi $\geq$ **0.95 precision** on all three registers (Devanagari, romanised, mixed)
- Survey: **adults 18+, anonymous**, no names/emails/identifiers, consent before the first question
- **Guide notified in writing 14 Aug 2026** — official PES account, all four of us on the mail, design disclosed, explicit "does the department need more approval?"

HARDEST QUESTIONS

"Did you take permission for the survey? Where is the ethics approval?"
Learn this one verbatim — §10 of the notes. There is a written record: 14 Aug 2026, official PES account, guide plus all teammates, full design disclosed, explicit approval query. That is faculty notification, **not** an ethics-committee determination, and no paper of ours says otherwise. **Concede the timing before they ask:** it went out while the form was still open — after the main wave. Notification should have preceded the first response; it did not, and we corrected course mid-window. **If asked "did she reply?"** — state only what the record documents and stop. We claim no reply, no approval, and no absence-of-objection.

"You alert on swear words but stay silent on self-harm."
Name the asymmetry first — it is real and we published it. It is a refusal to hard-code a contested clinical judgement into an automatic escalation. The tiered wellbeing-check is documented as a proposal and deliberately not shipped.

"Why only children? Why not adults?"
A guardian with a duty of care is what makes screening someone who has not asked for it both possible and proportionate. The same system pointed at an adult is either a self-help journal or spyware.

"You are recording a child's microphone."
Layered, and every layer is checkable: consent $\rightarrow$ on-device STT $\rightarrow$ server-side deletion $\rightarrow$ VAD gate $\rightarrow$ export and delete rights that share one scope.

You open the room — the framing you set in slides 2–3 is the frame they judge the rest by. **Hand-offs:** fairness *numbers* $\rightarrow$ Kaustubh (you own *why it matters*) · capture mechanics $\rightarrow$ Khushee.